¿Es fraude? — Judges' brief

A second opinion before you fall for a scam

A free, open-source web app for Mexico, in Spanish and English. It checks a suspicious message, screenshot, link, email, call or loan app against official Mexican sources, quotes every warning sign, never says "safe", and walks victims through reporting it. Built for the Nebius x NVIDIA Global AI Hackathon, Best Apps and Agents track.

Live demo (English)	**Code (MIT)**	**Demo video (2:14)**
es-fraude-seven.vercel.app/?lang=en	github.com/ramsesavi94-maker/es-fraude	youtu.be/5_hwRbQ4zE

Try it in two minutes (no account needed)

1. Open the English demo and tap **Fake tax authority (SAT)**. In about 6–9 s you get a high-risk verdict with quoted signs, the lookalike domain check and real official SAT alerts found by Tavily.
2. Scroll to **Report it, step by step**, tap **Yes**, choose **Ciudad de México**: verified urgent steps, a draft complaint built only in your browser, and one-tap routes (online prosecutor complaint, cyber police, 089 with a script).
3. Tap **Email "from SAT"** (a Gmail account posing as the tax authority) and **Real bank alert** (a legitimate message: no alarm, but never "safe").
4. Optional: the **Screenshot** tab reads a photo of an SMS with Nemotron 3.5 Super VL. A sample is in the repo: `samples/sms-falso-banco.png`.

Example messages are in Spanish, as people in Mexico receive them; with `?lang=en` the analysis is written in English while quotes stay verbatim. The public demo has a per-IP hourly limit and a daily cap to stay within free credits.

Why it matters

Fraud is the most frequent crime in Mexico: an estimated 8.1 million frauds in 2025, 80.6% online or electronic, and 96.5% never reported (UNODC-INEGI, ENVIPE 2026). Victims ask Facebook groups before official channels. ¿Es fraude? answers about the message in front of them and ends in a concrete, verified next step.

How NVIDIA Nemotron and Nebius Token Factory are used

1 · Read	**2 · Verify**	**3 · Reason**	**4 · Act**
Nemotron 3.5 Super VL Nemotron 3 Super Screenshot transcription; extraction of sender, links, phones, amounts, requests.	Deterministic hostname and lookalike checks, domain age (RDAP), Tavily searches limited to official Mexican domains.	Nemotron 3 Ultra Verdict against 12 scam patterns documented by Mexican authorities. JSON mode, temperature 0.	Steps, phone numbers, deadlines and reporting routes come only from a verified catalog.

All inference runs at request time on Nebius Token Factory (OpenAI-compatible API). We compared models on real calls: Super was faster (≈2 s vs ≈3.6 s) and more accurate than Nano for extraction, so we use Super there and reserve Ultra for reasoning. Measured cost: about US$0.017 per analysis.

Results (synthetic sets written from official alerts)

SET	CASES	SCAMS CAUGHT	LEGITIMATE, NO ALARM	NOTES
Third set	40	**25/25**	**15/15**	**Unseen, measured once with no changes**; 27 hard cases; also 25/25 and 15/15 with English answers
Links & emails	30	**18/18**	**12/12**	**Unseen**: lookalike domains, deceptive subdomains, Cyrillic letters, IP hosts, shorteners, free accounts posing as SAT/IMSS/BBVA
Holdout	40	**25/25**	**15/15**	First run before fixes: 24/25 and 14/15
Tuning set	45	**30/30**	**15/15**	Used to tune rules

Run any set with `npx tsx --env-file=.env scripts/evaluate.ts eval/third.json`. The sets are synthetic; testing with real users is the next step.

Safety and grounding decisions

- **No invented contacts.** Early on, Ultra invented a SAT phone number and a non-existent reporting short code. The model now only classifies and explains.
- **Never "safe".** Enforced in code with a fixed message when no signs are found.
- **Quotes are evidence.** Signals must quote the message or an official page; quotes of our own check text are dropped in code.
- **Relevant evidence only.** Search results must name the institution as a whole word and talk about fraud.
- **Privacy.** Nothing is stored; searches receive names, domains and phone numbers, never the message; complaint details stay in the browser.
- **Honest routes.** Each reporting channel says whether it is guidance, a claim or a formal criminal complaint.

Where to look in the repo

`src/analyze.ts`	Pipeline: read, extract, verify, reason, compose
`src/catalog.ts`	Verified patterns, channels, domains
`src/report.ts`	Reporting routes, urgent steps, evidence
`src/linkcheck.ts`	Link and email checks, RDAP
`src/llm.ts`	Token Factory client
`eval/`	Evaluation sets and results
`docs/research/`	Sourced research behind the catalog and routes

Known limitations

- Evaluation is synthetic; real-world accuracy is not yet measured.
- Domain age is unavailable for .mx (no public RDAP).
- Online complaint routes verified for CDMX, State of Mexico and Nuevo León; other states get in-person guidance and the official cyber-police directory.
- In the video, analysis waits are shortened and the narration is AI-generated.

Team: Nelson Ramses Aviles, with Claude Code (architecture and code) and Codex (research on Mexican scam patterns, official reporting routes and evaluation sets). This app is guidance, not legal advice.